Atomistic and Data-Driven Modeling of Materials for Energy Applications - Postdoctoral Researcher

Lawrence Livermore National Laboratory · Livermore, CA, United States

Job ID (this archive)	ENERGY-107
Employer	Lawrence Livermore National Laboratory
Location(s)	Livermore, CA, United States
Career stage (archive tag)	Graduate degree / research (Postdoctoral)
Location group	Other U.S.
Majors (archive tags)	Mechanical Engineering, Materials Science & Engineering, Data Science
Energy area	Energy materials / scientific ML
Pay listed	Not listed
Posted / listing date	46317.870481770835
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://jobs.smartrecruiters.com/LLNL/3743990016356547-atomistic-and-data-driven-modeling-of-materials-for-energy-applications-postdoctoral-researcher
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-078

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Job description

Company Description

Join us and make YOUR mark on the World!

Lawrence Livermore National Laboratory (LLNL) has turned bold ideas into world-changing impact advancing science and technology to strengthen U.S. security and promote global stability.

Our mission spans four critical national security areas nuclear deterrence, threat preparedness, energy security, and multi-domain defense empowering teams to take on the toughest challenges of today and tomorrow. With a culture built on innovation and operational excellence, LLNL is a place where your expertise can make a real impact.

Job Description

We have multiple openings for Postdoctoral Researcher Positions to conduct mentored research in atomistic and data-driven modeling of materials for energy applications. Key focus areas include investigating reactivity, transport, and phase evolution at heterogeneous interfaces; predicting materials degradation and coupled chemo-electro-mechanical response under operating conditions; understanding electronic properties of materials under non-equilibrium conditions; and developing data science approaches for predicting materials performance across scales. You will work closely with a multidisciplinary team in support of projects sponsored by the Basic Energy Sciences, Transportation Technologies Offices, and Office of Electricity within the Department of Energy, and internal LDRD programs. This position is within the Quantum Simulations Group within the Materials for Emerging Applications and Extreme Conditions section of the Materials Science Division in the Physical and Life Sciences Directorate.

This position requires full-time on-site presence due to the nature of the work.

Note: This is a two-year Postdoctoral appointment with the possibility of extension to a maximum of three years. Eligible candidates are recent PhDs within five years of the month of the degree award at time of hire date.

You will

- Perform electronic structure theory-based simulations of complex materials, such as oxides and multi-element systems.
- Perform thermodynamic and kinetic analyses of chemical reactions and phase transitions at surfaces and interfaces.
- Perform non-equilibrium molecular dynamics simulations & first-principles calculations to study materials response under externally applied stimuli.
- Derive structure-composition-property relationships using statistical, analytical, and machine-learning based methodologies.
- Develop machine-learning interatomic potentials and surrogate models based on physics-informed descriptors for property evaluation and prediction.
- Contribute to the planning, design, and execution of assigned research activities under the guidance of senior scientists and project leadership. Collaborate with computational and experimental scientists in a multidisciplinary team environment to accomplish research goals.
- Develop increasing technical independence while pursuing research activities aligned with project and program goals; interact with collaborators within and outside the Laboratory. Document research, contribute to peer-reviewed publications, and present results within the DOE community and at conferences/technical meetings.
- Perform other duties as assigned.

Qualifications

- Must be eligible to access the Laboratory in compliance with Section 3112 of the National Defense Authorization Act (NDAA). See Additional Information section below for details.
- Ph.D. degree completed, or anticipated to be completed by the hire date, in Materials Science, Chemistry, Physics, Mechanical Engineering or a related field.
- Experience in the application of density functional theory and/or advanced electronic structure theory for simulating complex materials, including oxides and multi-element systems
- Experience performing large-scale molecular dynamics simulations on high-performance computing environments.
- Additional experience in at least one of the following methods: machine-learning based approaches, cluster expansion, advanced statistical analysis, kinetic Monte Carlo simulations, phase-field modeling, reduced order, descriptor or surrogate model development as relevant to energy related applications.
- Ability to perform technical assignments with increasing independence, analyze results, and contribute to solutions for defined research problems. Ability to contribute to research directions under mentorship and communicate results effectively through peer-reviewed publications and technical presentations. Proficient verbal and written communication skills to collaborate effectively in a team environment, prepare written reports and present and explain technical information.
- Interpersonal skills necessary to interact with scientists, engineers and other technical and administrative staff in a collaborative, multidisciplinary team environment.

Qualifications We Desire

- Experience with computational workflow development and multiscale model integration
- Experience in modeling highly disordered materials and/or heterogeneous interfaces
- Experience with SI (Super Intelligence, formerly referred to as Artificial Intelligence, or AI)/ML-enabled materials discovery, surrogate modeling, and generative models.
- Experience with Bayesian optimization, uncertainty quantification, sensitivity analysis, and Pareto front analysis.
- Experience supporting DOE consortia or multi-laboratory data integration efforts.

Pay Range

$123,048 Annually

This is the lowest to highest salary we in good faith believe we would pay for this role at the time of this posting. An employee's position within the salary range will be based on several factors including, but not limited to, specific competencies, relevant education, qualifications, certifications, experience, skills, seniority, geographic location, performance, and business or organizational needs.

Additional Information

#LI-Onsite

Position Information

This is a Postdoctoral appointment with the possibility of extension to a maximum of three years. The postdoctoral appointment is a temporary, mentored training position intended for individuals in the early stages of their postdoctoral research careers. Consistent with the structure of the program, applicants must have received their Ph.D. (or equivalent doctorate) within five years of the anticipated start date.

Why Lawrence Livermore National Laboratory?

- Included in 2026 Best Places to Work by Glassdoor!
- Flexible Benefits Package
- 401(k)
- Relocation Assistance
- Education Reimbursement Program
- Flexible schedules (*depending on project needs)
- Our values - visit https://www.llnl.gov/inclusion/our-values

Security Clearance

None required. However, if your assignment is longer than 179 days cumulatively within a calendar year, you must go through the Personal Identity Verification process. This process includes completing an online background investigation form and receiving approval of the background check.

National Defense Authorization Act (NDAA)

The 2025 National Defense Authorization Act (NDAA), Section 3112, generally prohibits citizens of China, Russia, Iran and North Korea without dual US citizenship or legal permanent residence from accessing specific non-public areas of national security or nuclear weapons facilities. The restrictions of NDAA Section 3112 apply to this position. To be qualified for this position, Candidates must be eligible to access the Laboratory in compliance with Section 3112.

Pre-Employment Drug Test

External applicant(s) selected for this position must pass a post-offer, pre-employment drug test. This includes testing for use of marijuana as Federal Law applies to us as a Federal Contractor.

Wireless and Medical Devices

Per the Department of Energy (DOE), Lawrence Livermore National Laboratory must meet certain restrictions with the use and/or possession of mobile devices in Limited Areas. Depending on your job duties, you may be required to work in a Limited Area where you are not permitted to have a personal and/or laboratory mobile device in your possession. This includes, but not limited to cell phones, tablets, fitness devices, wireless headphones, and other Bluetooth/wireless enabled devices.

If you use a medical device, which pairs with a mobile device, you must still follow the rules concerning the mobile device in individual sections within Limited Areas. Sensitive Compartmented Information Facilities require separate approval. Hearing aids without wireless capabilities or wireless that has been disabled are allowed in Limited Areas, Secure Space and Transit/Buffer Space within buildings.

How to identify fake job advertisements

Please be aware of recruitment scams where people or entities are misusing the name of Lawrence Livermore National Laboratory (LLNL) to post fake job advertisements. LLNL never extends an offer without a personal interview and will never charge a fee for joining our company. All current job openings are displayed on the Career Page under “Find Your Job” of our website. If you have encountered a job posting or have been approached with a job offer that you suspect may be fraudulent, we strongly recommend you do not respond.

To learn more about recruitment scams:
https://www.llnl.gov/sites/www/files/2023-05/LLNL-Job-Fraud-Statement-Updated-4.26.23.pdf

Equal Employment Opportunity

We are an equal opportunity employer that is committed to providing all with a work environment free of discrimination and harassment. All qualified applicants will receive consideration for employment without regard to race, color, religion, marital status, national origin, ancestry, sex, sexual orientation, gender identity, disability, medical condition, pregnancy, protected veteran status, age, citizenship, or any other characteristic protected by applicable laws.

Reasonable Accommodation

Our goal is to create an accessible and inclusive experience for all candidates applying and interviewing at the Laboratory. If you need a reasonable accommodation during the application or the recruiting process, please use our online form to submit a request.

California Privacy Notice

The California Consumer Privacy Act (CCPA) grants privacy rights to all California residents. The law also entitles job applicants, employees, and non-employee workers to be notified of what personal information LLNL collects and for what purpose. The Employee Privacy Notice can be accessed here.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official SmartRecruiters job data reports active=true and supplies a job-specific apply URL. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Note: This is a two-year Postdoctoral appointment with the possibility of extension to a maximum of three years. Eligible candidates are recent PhDs within five years of the month of the degree award at time of hire date. | Ph.D. degree completed, or anticipated to be completed by the hire date, in Materials Science, Chemistry, Physics, Mechanical Engineering or a related field. | This is a Postdoctoral appointment with the possibility of extension to a maximum of three years. The postdoctoral appointment is a temporary, mentored training position intended for individuals in the early stages of their postdoctoral research careers. Consistent with the structure of the program, applicants must have received their Ph.D. (or equivalent doctorate) within five years of the anticipated start date.

Key skills listed in the archive: optimization

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer SmartRecruiters Employer / ATS identifier: REF8671X

Career stage: Postdoctoral

Major fit: Advanced research pathway: data-driven materials modeling; relevant PhD and physics/materials expertise required. Energy-sector connection: Atomistic simulation and machine-learning models of materials for energy applications.

Retrieval note: New posting found in expansion round. Official SmartRecruiters job data reports active=true and supplies a job-specific apply URL. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer SmartRecruiters. Source identifier: REF8671X.

Sources and retrieval details

Employer posting

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Retrieved source

https://api.smartrecruiters.com/v1/companies/LLNL/postings/3743990016356547

Retrieved: 2026-10-26T16:35:32.639669+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.